

$$\begin{aligned}
 1) \lim_{k \rightarrow \infty} 1 &= 1 & 2) \lim_{k \rightarrow \infty} 2 + \frac{1}{k} &= 2 + \frac{1}{\infty} = 2 & 3) \lim_{k \rightarrow \infty} \frac{3}{k^2} &= \frac{3}{\infty} = 0 \\
 4) \lim_{k \rightarrow \infty} e^{-k} \sin k &= \lim_{k \rightarrow \infty} \frac{\sin k}{e^k} = \frac{\sin \infty}{\infty} = \frac{c}{\infty} = 0
 \end{aligned}$$

$e^\infty = \infty$

Example 3 Show that

$$\mathbf{x}^{(k)} = (x_1^{(k)}, x_2^{(k)}, x_3^{(k)}, x_4^{(k)})^t = \left(1, 2 + \frac{1}{k}, \frac{3}{k^2}, e^{-k} \sin k \right)^t.$$

converges to $\mathbf{x} = (1, 2, 0, 0)^t$ with respect to the l_∞ norm.

Solution Because

$$\lim_{k \rightarrow \infty} 1 = 1, \quad \lim_{k \rightarrow \infty} (2 + 1/k) = 2, \quad \lim_{k \rightarrow \infty} 3/k^2 = 0 \quad \text{and} \quad \lim_{k \rightarrow \infty} e^{-k} \sin k = 0,$$

Theorem 7.6 implies that the sequence $\{\mathbf{x}^{(k)}\}$ converges to $(1, 2, 0, 0)^t$ with respect to the l_∞ norm. ■

Example 5 Determine $\|A\|_\infty$ for the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & -1 \\ 5 & -1 & 1 \end{bmatrix}.$$

Norm of matrix in $L_\infty \rightarrow \|A\|_\infty$
 ① add each row
 ② take max

Solution We have

$$\sum_{j=1}^3 |a_{1j}| = |1| + |2| + |-1| = 4, \quad \sum_{j=1}^3 |a_{2j}| = |0| + |3| + |-1| = 4,$$

$$\sum_{j=1}^3 |a_{3j}| = |5| + |-1| + |1| = 7, \quad \|A\|_\infty = \max\{4, 4, 7\} = 7$$

and

$$\sum_{j=1}^3 |a_{3j}| = |5| + |-1| + |1| = 7.$$

So, Theorem 7.11 implies that $\|A\|_\infty = \max\{4, 4, 7\} = 7$.

In the next section, we will discover an alternative method for finding the l_2 norm of a matrix.

Norm of vector = length

$$\|x\| = 0$$

$$x = (x_1, x_2) \rightarrow \|x\| = \sqrt{(x_1)^2 + (x_2)^2}$$

$$x = (x_1, x_2, x_3, x_4) \rightarrow \|x\| = \sqrt{(x_1)^2 + (x_2)^2 + (x_3)^2 + (x_4)^2}$$

$$\|cx\| = |c| \|x\|$$

Scalar
constant

$$\|x+y\| \leq \|x\| + \|y\|$$

$$\|x\|_\infty = \max |x_i|$$

in finite space

$$x = (2, -3, 5, -9) \rightarrow \|x\|_\infty = \max\{|2|, |-3|, |5|, |-9|\} = 9$$

$$\|x\|_2$$

finite space

Distance in L_2

$$\|x-y\|_2 = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \dots + (x_n-y_n)^2}$$

Distance in L_∞

$$\|x-y\|_\infty = \max\{|x_i - y_i|\}$$

EXERCISE SET 7.1

1. Find l_∞ and l_2 norms of the vectors.

1) a) $\|x\|_\infty = \max\{|3|, |-4|, |0|, |\frac{3}{2}|\} = 4$

$$\|x\|_2 = \sqrt{3^2 + (-4)^2 + 0^2 + (\frac{3}{2})^2} = 5.22$$

1) c) $\|x\|_\infty = \max\{| \sin k |, | \cos k |, | 2^k | \} = 2^k$

$$\|x\|_2 = \sqrt{(\sin k)^2 + (\cos k)^2 + (2^k)^2} = \sqrt{1 + 2^{2k}}$$

2. Find l_∞ and l_2 norms of the vectors.

a. $x = (2, -2, 1)^t$

b. $x = (-4/5, -2/5, 1/5, 2/5)^t$

c. $x = ((2+k)/k, 1/\sqrt{k}, -3)^t$ for a fixed positive integer k

d. $x = ((3k+1)/(2k), 2, 0, 1/k)^t$ for a fixed positive integer k

3. Prove that the following sequences are convergent and find their limits.

a. $x^{(k)} = (1/k, e^{1-k}, -2/k^2)^t$

b. $x^{(k)} = (e^{-k} \cos k, k \sin(1/k), 3 + k^{-2})^t$

c. $x^{(k)} = (ke^{-k^2}, (\cos k)/k, \sqrt{k^2 + k} - k)^t$

d. $x^{(k)} = (e^{1/k}, (k^2 + 1)/(1 - k^2), (1/k^2)(1 + 3 + 5 + \dots + (2k - 1)))^t$

4. Prove that the following sequences are convergent and find their limits.

a. $x^{(k)} = (2 + 1/k, -2 + 1/k, 1 + 1/k^2)^t$

b. $x^{(k)} = ((2+k)/k, k/(2+k), (2k+1)/k)^t$

c. $x^{(k)} = ((3k+1)/k^2, (1/k) \ln k, k^2 e^{-k}, 2k/(1+2k))^t$

d. $x^{(k)} = \left(\frac{\cos k}{k}, \frac{\sin k}{k}, \frac{1-k}{k^2+1}, \frac{3k-2}{4k+1} \right)^t$

5. Find the l_∞ norm of the matrices.

a.
$$\begin{bmatrix} 10 & 15 \\ 0 & 1 \end{bmatrix}$$

$$a_{1j} = |10| + |15| = 25$$

$$a_{2j} = |0| + |1| = 1$$

$$\|A\|_\infty = \max\{25, 1\} = 25$$

c.
$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

b.
$$\begin{bmatrix} 10 & 0 \\ 15 & 1 \end{bmatrix}$$

d.
$$\begin{bmatrix} 4 & -1 & 7 \\ -1 & 4 & 0 \\ -7 & 0 & 4 \end{bmatrix}$$

6. Find the l_∞ norm of the matrices.

a.
$$\begin{bmatrix} 10 & -1 \\ -1 & 11 \end{bmatrix}$$

c.
$$\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 1 \\ 1 & 0 & 0 \\ \pi & -1 & 2 \end{bmatrix}$$

b.
$$\begin{bmatrix} 11 & -11 \\ 0 & 1 \end{bmatrix}$$

d.
$$\begin{bmatrix} 1/3 & -1/3 & 1/3 \\ -1/4 & 1/2 & 1/4 \\ 2 & -2 & -1 \end{bmatrix}$$

7. The following linear systems $A\mathbf{x} = \mathbf{b}$ have $\mathbf{x}$ as the actual solution and $\tilde{\mathbf{x}}$ as an approximate solution. Compute $\|\mathbf{x} - \tilde{\mathbf{x}}\|_\infty$ and $\|A\tilde{\mathbf{x}} - \mathbf{b}\|_\infty$.

a.
$$\frac{1}{2}x_1 + \frac{1}{3}x_2 = \frac{1}{63},$$

$$\frac{1}{3}x_1 + \frac{1}{4}x_2 = \frac{1}{168},$$

$$\mathbf{x} = \left(\frac{1}{7}, -\frac{1}{6}\right)^t,$$

$$\tilde{\mathbf{x}} = (0.142, -0.166)^t.$$

b.
$$x_1 + 2x_2 + 3x_3 = 1,$$

$$2x_1 + 3x_2 + 4x_3 = -1,$$

$$3x_1 + 4x_2 + 6x_3 = 2,$$

$$\mathbf{x} = (0, -7, 5)^t,$$

$$\tilde{\mathbf{x}} = (-0.33, -7.9, 5.8)^t.$$

c.
$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 1, \\ 2x_1 + 3x_2 + 4x_3 &= -1, \\ 3x_1 + 4x_2 + 6x_3 &= 2, \\ \mathbf{x} &= (0, -7, 5)^t, \\ \tilde{\mathbf{x}} &= (-0.2, -7.5, 5.4)^t. \end{aligned}$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

d.
$$\begin{aligned} 0.04x_1 + 0.01x_2 - 0.01x_3 &= 0.06, \\ 0.2x_1 + 0.5x_2 - 0.2x_3 &= 0.3, \\ x_1 + 2x_2 + 4x_3 &= 11, \\ \mathbf{x} &= (1.827586, 0.6551724, 1.965517)^t, \\ \tilde{\mathbf{x}} &= (1.8, 0.64, 1.9)^t. \end{aligned}$$

8. The following linear systems $A\mathbf{x} = \mathbf{b}$ have $\mathbf{x}$ as the actual solution and $\tilde{\mathbf{x}}$ as an approximate solution. Compute $\|\mathbf{x} - \tilde{\mathbf{x}}\|_\infty$ and $\|A\tilde{\mathbf{x}} - \mathbf{b}\|_\infty$.

a.
$$\begin{aligned} 3.9x_1 + 1.5x_2 &= 5.4, \\ 6.8x_1 - 2.9x_2 &= 3.9, \\ \mathbf{x} &= (1, 1)^t, \\ \tilde{\mathbf{x}} &= (0.98, 1.02)^t. \end{aligned}$$

b.
$$\begin{aligned} x_1 + 2x_2 &= 3, \\ 1.001x_1 - x_2 &= 0.001, \\ \mathbf{x} &= (1, 1)^t, \\ \tilde{\mathbf{x}} &= (1.02, 0.98)^t. \end{aligned}$$

c.
$$\begin{aligned} x_1 + x_2 + x_3 &= 2\pi, \\ -x_1 + x_2 - x_3 &= 0, \\ x_1 + x_3 &= \pi, \\ \mathbf{x} &= (0, \pi, \pi)^t, \\ \tilde{\mathbf{x}} &= (0.1, 3.18, 3.10)^t. \end{aligned}$$

d.
$$\begin{aligned} 0.04x_1 + 0.01x_2 - 0.01x_3 &= 0.0478, \\ 0.4x_1 + 0.1x_2 - 0.2x_3 &= 0.413, \\ x_1 + 2x_2 + 3x_3 &= 0.14, \\ \mathbf{x} &= (1.81, -1.81, 0.65)^t, \\ \tilde{\mathbf{x}} &= (2, -2, 1)^t. \end{aligned}$$

THEORETICAL EXERCISES

9. a. Verify that the function $\|\cdot\|_1$, defined on $\mathbb{R}^n$ by

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|,$$

is a norm on $\mathbb{R}^n$.

- b. Find $\|\mathbf{x}\|_1$ for the vectors given in Exercise 1.
 c. Prove that for all $\mathbf{x} \in \mathbb{R}^n$, $\|\mathbf{x}\|_1 \geq \|\mathbf{x}\|_2$.

d. $\mathbf{x}^{(k)} = \left(\frac{\cos k}{k}, \frac{\sin k}{k}, \frac{1-k}{k^2+1}, \frac{3k-2}{4k+1} \right)^T$

5. Find the l_∞ norm of the matrices.

a. $\begin{bmatrix} 10 & 15 \\ 0 & 1 \end{bmatrix}$
 $a_{1j} = |10| + |15| = 25$
 $a_{2j} = |0| + |1| = 1$
 $\|A\|_\infty = \max\{25, 1\} = 25$

c. $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$
 Solved one it above the set

b. $\begin{bmatrix} 10 & 0 \\ 15 & 1 \end{bmatrix}$
 d. $\begin{bmatrix} 4 & -1 & 7 \\ -1 & 4 & 0 \\ -7 & 0 & 4 \end{bmatrix}$

6. Find the l_∞ norm of the matrices.

a. $\begin{bmatrix} 10 & -1 \\ -1 & 11 \end{bmatrix}$

b. $\begin{bmatrix} 11 & -11 \\ 0 & 1 \end{bmatrix}$

c. $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 1 \\ 1 & 0 & 0 \\ \pi & -1 & 2 \end{bmatrix}$

d. $\begin{bmatrix} 1/3 & -1/3 & 1/3 \\ -1/4 & 1/2 & 1/4 \\ 2 & -2 & -1 \end{bmatrix}$

7. The following linear systems $Ax = b$ have x as the actual solution and $\tilde{x}$ as an approximate solution. Compute $\|x - \tilde{x}\|_\infty$ and $\|A\tilde{x} - b\|_\infty$.

a. $\frac{1}{2}x_1 + \frac{1}{3}x_2 = \frac{1}{63}$,
 $\frac{1}{3}x_1 + \frac{1}{4}x_2 = \frac{1}{168}$,
 $x = \left(\frac{1}{7}, -\frac{1}{6}\right)^t$

b. $x_1 + 2x_2 + 3x_3 = 1$,
 $2x_1 + 3x_2 + 4x_3 = -1$,
 $3x_1 + 4x_2 + 6x_3 = 2$,
 $x = (0, -7, 5)^t$,
 $\tilde{x} = (-0.33, -7.9, 5.8)^t$

c. $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 2x_1 + 3x_2 + 4x_3 = -1 \\ 3x_1 + 4x_2 + 6x_3 = 2 \end{cases}$
 $\tilde{x} = (0.142, -0.166)^t$
 $x = (0, -7, 5)^t$, exact
 $\tilde{x} = (-0.2, -7.5, 5.4)^t$, approx

d. $0.04x_1 + 0.01x_2 - 0.01x_3 = 0.06$,
 $0.2x_1 + 0.5x_2 - 0.2x_3 = 0.3$,
 $x_1 + 2x_2 + 4x_3 = 11$,
 $x = (1.827586, 0.6551724, 1.965517)^t$,
 $\tilde{x} = (1.8, 0.64, 1.9)^t$

8. The following linear systems $Ax = b$ have x as the actual solution and $\tilde{x}$ as an approximate solution. Compute $\|x - \tilde{x}\|_\infty$ and $\|A\tilde{x} - b\|_\infty$.

7C $x = (0, -7, 5)^t$
 $\tilde{x} = (-0.2, -7.5, 5.4)^t$
 $\|x - \tilde{x}\|_\infty = \max\{|0 - (-0.2)|, |-7 - (-7.5)|, |5 - 5.4|\}$
 $\max\{0.2, 0.5, 0.4\} = 0.5$

$\|A\tilde{x} - b\|_\infty$
 $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{bmatrix}$, $b = \begin{bmatrix} -0.2 \\ -1.3 \\ 1.8 \end{bmatrix}$
 $b - \tilde{b}$

$\|A\tilde{x} - b\|_\infty = \max\{|1 - 1|, |1 - 1.3 + 1|, |1.8 - 2|\}$
 $\max\{0, 0.3, 0.2\} = 0.3$

Example 3 Show that

$$\lim_{k \rightarrow \infty} 2 + \frac{1}{k} = 2 + \frac{1}{\infty} = 2$$

$$\mathbf{x}^{(k)} = (x_1^{(k)}, x_2^{(k)}, x_3^{(k)}, x_4^{(k)})^t = \left(1, 2 + \frac{1}{k}, \frac{3}{k^2}, e^{-k} \sin k \right)^t$$

$\lim$

converges to $\mathbf{x} = (1, 2, 0, 0)^t$ with respect to the l_∞ norm.

Solution Because

$$\lim_{k \rightarrow \infty} 1 = 1, \quad \lim_{k \rightarrow \infty} (2 + 1/k) = 2, \quad \lim_{k \rightarrow \infty} 3/k^2 = 0 \quad \text{and} \quad \lim_{k \rightarrow \infty} e^{-k} \sin k = 0,$$

Theorem 7.6 implies that the sequence $\{\mathbf{x}^{(k)}\}$ converges to $(1, 2, 0, 0)^t$ with respect to the l_∞ norm. ■

$$\lim_{k \rightarrow \infty} 1 = 1 \checkmark$$

$$\lim_{k \rightarrow \infty} 2 + \frac{1}{k} = 2 \checkmark$$

$$\lim_{k \rightarrow \infty} \frac{3}{k^2} = 0 \checkmark$$

$$\lim_{k \rightarrow \infty} e^{-k} \sin k = \lim_{k \rightarrow \infty} \frac{\sin k}{e^k} = 0 \checkmark$$

(0,1)
max: 1

max go to infinity

transpose

$e^\infty = \infty$

$$\lim_{k \rightarrow \infty} e^{-k} \sin k = \frac{\sin k}{e^k}$$

$$= \frac{\sin \infty}{\infty}$$

$$= \frac{\sin \infty}{\infty}$$

$$\lim_{k \rightarrow \infty} \frac{3}{k^2} = \frac{3}{\infty^2} = 0$$